

On the Non-Identifiability of Steering Vectors in Large Language Models

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THE CENTRAL CLAIM

Activation steering controls LLM behavior by adding a direction extracted from contrasting prompts — and is routinely read as evidence that a concept *lives along* that direction. That reading assumes the direction is uniquely recoverable from behavior.

It is not. An entire high-dimensional **family of directions** produces behavior indistinguishable from the original.

Row space

OBSERVABLE

$$Jv \neq 0$$

The only component steering can move. Everything you measure at the output lives here.

Null space

UNOBSERVABLE

$$Jv_0 = 0$$

Add any direction from here and not a single output changes. Infinitely many of them exist.

FOUR FINDINGS

1 Orthogonal twins are invisible

Random directions orthogonal to v leave behavior unchanged — mean Cohen's d of **0.119** (Llama) and **0.131** (Qwen).

2 Different methods, same behavior

Vectors from **cos −0.54** to **+0.32** — near-orthogonal — yet every pair stays at $d < 0.27$.

3 The class is high-dimensional

50 orthogonal directions per trait all land near $|d| \approx 0.08$ – 0.13 — not a thin shell around v .

4 Stable, not an artifact

Holds across steering strength $\alpha \in [0, 3]$ and survives topic- and genre-shift — a geometric property.

EXPERIMENTAL DESIGN

LLAMA-3.1 8B · MHA · L16/32

QWEN2.5 3B · GQA · L18/36

5 traits (formality, politeness, sentiment, truthfulness, agreeableness) · **50** contrastive prompt pairs each · effect measured as **Cohen's d** via pretrained classifiers · tests: orthogonal perturbation, method comparison, null-space spanning, multi-environment.

PERTURBATIONS STAY NEGLIGIBLE

Mean Cohen's d , v vs $v + v_\perp$ · $n = 10$ seeds

TRAIT	LLAMA-3.1 8B	QWEN2.5 3B
Formality	0.059	0.151
Politeness	0.159	0.120
Sentiment	0.104	0.150
Truthfulness	0.085	0.092
Agreeableness	0.186	0.156

— Vertical tick marks the **$|d| = 0.2$** negligible-effect threshold — all ten cells fall below it.

WHY DOES THIS HAPPEN?

The map from activations to logits, $J = \partial o / \partial h$, is rank-deficient in overparameterized models. Its null space has dimension $d - \text{rank}(J) \geq 1$, so infinitely many directions satisfy $Jv_0 = 0$.

Behavior pins down only the row-space shadow of v — never v itself.

$$\begin{aligned} J(v + v_0) &= Jv + Jv_0 \\ &= Jv + 0 \end{aligned}$$

Identical logits, identical behavior — for any null-space direction v_0 .

Behavioral equivalence cannot certify what a steering vector represents.

Reliable interpretability and alignment need structural constraints beyond output behavior — the full proof, code and experiments are in the paper.

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